Maisha Zaman Khan

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Summary

Computer Science and Engineering graduate with a strong foundation in analytical thinking, data analysis, and technology-driven problem-solving.Demonstrates the ability to translate complex challenges into practical solutions through software development, machine learning, and research. Combines technical proficiency with attention to detail, adaptability, and a commitment to continuous learning. Focused on improving processes, supporting informed decision-making, and delivering accurate, efficient, and results-driven outcomes through a structured, analytical, and solution-oriented approach.

Education

2022 – 2026 **BSc in Computer Science and Engineering**, *BRAC University, Dhaka*, *CGPA: 3.09 / 4.00*

2020 **HSC**, *Dhaka City College, Dhaka*, *GPA: 5.00 / 5.00*

2018 **SSC**, *Udayan Higher Secondary School and College, Dhaka*, *GPA: 5.00 / 5.00*

Research

2026 **Deep Learning for Early Alzheimer's Detection: Advancing Diagnosis with AI-Powered Models**, *Under Review, IEEE WEICON-ECE 2026*

- Designed two novel architectures to classify five Alzheimer's stages from brain MRI: **MADnet**, a 0.0986M-parameter lightweight CNN with a multi-scale stem and SE blocks (94.58% accuracy, 92.29% macro F1); and **CATnet**, a 0.565M-parameter hybrid CNN-Transformer using cross-attention tokenization (94.24% accuracy, 91.60% macro F1) — both outperforming much larger pretrained baselines such as VGG19 (139.6M params). Used a training-only class-balancing pipeline and Grad-CAM analysis to confirm model attention aligns with clinically relevant brain regions.

Projects

NeuralNest, Online Learning System, *Full-Stack Web Application | Python, Flask, MySQL, HTML, CSS, JavaScript, Matplotlib* *GitHub*

- Developed a full-stack online learning platform with separate functionality for students, instructors, and admins using Flask, MySQL, and HTML/CSS/JavaScript.
- Built a Flask backend with MySQL for course management, enrollment, PDF assignment submission with late detection, grading, and feedback, plus auto-graded MCQ exams, an 80% completion threshold, per-course leaderboards, certificate generation, and admin analytics charts.
- Implemented searchable and filterable course discovery, module-based course content access, and student profile updates to support the end-to-end learning workflow.

GNN-BERT for Music Context Understanding, *Neural Networks Course Project | Python, PyTorch, PyTorch Geometric, Transformers, librosa* *GitHub*

- Built a hybrid DistilBERT + GraphSAGE system for multi-label music tagging on GTZAN, fusing text embeddings with audio-segment graphs (temporal and chroma-similarity edges) to reach 0.82 macro-F1 and 0.90 AUC-PR, versus 0.60 macro-F1 for a mel-spectrogram CNN baseline.
- Engineered segment-level audio graphs using chroma and MFCC features, connecting segments through temporal and chroma-similarity edges to represent musical structure for GraphSAGE-based learning.

Stroke Prediction, Machine Learning Project, *Python, scikit-learn, Pandas, Matplotlib, Seaborn GitHub*

- Built a stroke-prediction pipeline on 5,110 patient records, cleaning and scaling the data, then comparing KNN, a neural network (MLP), and Logistic Regression alongside K-Means clustering using accuracy, recall, ROC-AUC, and confusion matrices, ranking models by AUC and stroke sensitivity to handle the class imbalance.

Skills

Technical Skills: Microsoft Excel, Google Sheets, Microsoft Word, PowerPoint, SQL, Python, C/C++, JavaScript, HTML/CSS, Data Analysis, Data Visualization, Data Validation, Report Preparation, NumPy, Pandas, Git, Linux, PyTorch, TensorFlow, OCR, CNNs, Grad-CAM

Soft Skills: Analytical Thinking, Problem-Solving, Attention to Detail, Time Management, Teamwork and Collaboration, Effective Communication, Adaptability